

# Outcomes, Events, and Probability

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## 1 Introduction

To compile this information I am using the book **A Modern Introduction to Probability and Statistics** by F.M. Dekking, C. Kraaikamp, H.P. Lophuä, and L.E. Meester. It can be found here: <https://link.springer.com/book/10.1007/1-84628-168-7>.

This document is designed to review the Probability and Statistics principles covered in the aforementioned book. I will create other “Guides” for additional topics. They can be found at: <https://github.com/KovasMcCann/Writings>.

## 2 Sample Spaces

A **sample space** is the set of all possible outcomes of an experiment. Each element in the sample space represents a potential result of the experiment we are interested in.

For example, the sample space for a single coin toss is:

$$\Omega = \{H, T\}$$

where  $H$  represents heads and  $T$  represents tails.

## 3 Events

An **event** is a subset of the sample space. It consists of one or more outcomes from the sample space that share a certain property of interest.

## 4 Probability

A **probability** is how likely an event is to occur.

The formal definition:

A probability function  $P$  on a finite sample space  $\Omega$  assigns to each event  $A$  in  $\Omega$  a number  $P(A)$  in  $[0, 1]$  such that:

- (i)  $P(\Omega) = 1$ , and
- (ii)  $P(A \cup B) = P(A) + P(B)$  if  $A$  and  $B$  are disjoint.

The number  $P(A)$  is called the probability that  $A$  occurs.

For example, in a coin flip the probability is:

$$P(H) = P(T) = \frac{1}{2}$$

where  $P(H)$  is the probability of getting heads and  $P(T)$  is the probability of getting tails.

## 5 Addition Rule

The **addition rule** helps us find the probability that at least one of two events occurs.

- **For disjoint events:**

$$P(A \cup B) = P(A) + P(B)$$

- **General case:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The subtraction of  $P(A \cap B)$  ensures that outcomes common to both  $A$  and  $B$  are not counted twice.

## 6 Multiplication Rule

The **multiplication rule** helps us find the probability that both of two events occur.

- **For independent events:**

$$P(A \cap B) = P(A) \cdot P(B)$$

- **General case:**

$$P(A \cap B) = P(A) \cdot P(B|A)$$

The conditional probability  $P(B|A)$  is the probability of  $B$  given that  $A$  has occurred.

## 7 Complement Rule

The **complement rule** states that the probability of an event not occurring is equal to one minus the probability of the event occurring.

If  $A$  is an event, then the complement of  $A$ , denoted  $A^c$ , is the event that  $A$  does not occur. The complement rule is given by:

$$P(A^c) = 1 - P(A)$$

## 8 Products of Sample Spaces

When considering multiple experiments, the sample space of the combined experiment is formed by the **Cartesian product** of the individual sample spaces.

- **Independent events:** If there are  $n$  independent experiments, each with the same sample space  $S$ , then the total number of possible outcomes is

$$|S|^n$$

where  $|S|$  is the number of outcomes in  $S$ .

- **General case:** If the experiments have sample spaces  $S_1, S_2, \dots, S_n$ , then the combined sample space is

$$S_1 \times S_2 \times \cdots \times S_n$$

and the total number of possible outcomes is

$$|S_1| \times |S_2| \times \cdots \times |S_n|$$

For dependent events, the sample space is still the Cartesian product, but probabilities may depend on previous outcomes (i.e., involve conditional probabilities).

## 9 Intersection of Events and the Minimum Rule

The probability of the intersection of two events  $A$  and  $B$ , denoted  $P(A \cap B)$ , represents the likelihood that both events occur.

It is **not generally true** that:

$$P(A \cap B) = \min(P(A), P(B))$$

This only holds when one event is a subset of the other (i.e.,  $A \subseteq B$  or  $B \subseteq A$ ).

**Counterexample:** Suppose  $P(A) = 0.6$ ,  $P(B) = 0.5$ , and  $A$  and  $B$  are independent. Then:

$$P(A \cap B) = P(A) \cdot P(B) = 0.6 \times 0.5 = 0.3$$

but

$$\min(P(A), P(B)) = 0.5$$

Clearly,

$$P(A \cap B) < \min(P(A), P(B))$$

**General Rule:** For any two events  $A$  and  $B$ , the following inequality always holds:

$$P(A \cap B) \leq \min(P(A), P(B))$$

This makes intuitive sense: the probability that both events happen cannot exceed the probability of either event alone.

## 10 Infinite Sample Spaces

An **infinite sample space** is one that contains an infinite number of outcomes. This can occur in situations involving continuous variables or repeated processes.

**Example: Infinite Dice Rolls** Consider the sample space of all possible outcomes when rolling a fair die infinitely many times. The sample space can be written as:

$$\Omega = \{(x_1, x_2, x_3, \dots) \mid x_i \in \{1, 2, 3, 4, 5, 6\} \text{ for } i = 1, 2, 3, \dots\}$$

Here, each outcome is an infinite sequence of die rolls. Since there are infinitely many positions and each can take on one of six values, the set of all such infinite sequences is **uncountably infinite**, similar to the cardinality of the real numbers between 0 and 1.